Ethics by Design: The FairScore Underwriting Case

Background

You are the newly appointed **Ethics Lead** at **Halcyon Health Insurance**, a mid-sized insurer rolling out **FairScore** — an AI underwriting system that sets individual premiums and flags claims for automatic denial based on a predictive risk score. FairScore ingests medical history, prescription records, wearable-device data (for customers who opt into a "wellness discount" program), and zip-code-level socioeconomic data to estimate an applicant's future healthcare costs.

The CFO is enthusiastic: FairScore has cut claims-processing costs by 30% in the pilot region and let Halcyon offer lower premiums to "low-risk" customers, a strong selling point against competitors. But early data shows something uncomfortable — FairScore assigns systematically higher premiums and more claim denials to customers in lower-income zip codes and to people with chronic conditions like diabetes and depression, even when their actual medical costs don't justify it. The wellness-tracker discount, meanwhile, is popular with younger, healthier customers, but almost unused by older customers and those without smartphones — deepening the gap.

Leadership wants FairScore live company-wide next quarter. You've been asked to weigh the decision, build the guardrails, and be ready to say who is accountable if the system gets it wrong for a real customer.

Part 1: Weighing the Decision — Utilitarian Analysis

1. Apply the **three-step utilitarian method** to Halcyon's decision to expand FairScore company-wide:
    - **Identify the possible actions** (e.g., full rollout, rollout with a human-review requirement for denials, rollout limited to pricing only with no automatic denials, no rollout).
    - **Estimate the benefits and harms** of each — for shareholders, for "low-risk" customers, and for customers with chronic conditions or in lower-income zip codes.
    - **Choose the action** you think produces the greatest net benefit, and justify it.
2. As FairScore scales from a pilot region to the whole company, how might **diminishing marginal utility** apply? Does every additional data point (e.g., adding zip-code data on top of medical history) keep producing proportionally better risk predictions, or does it start producing more harm than insight?
3. Whose "common good" is FairScore really serving — the customer pool as a whole, shareholders, or "low-risk" customers specifically? Construct a scenario where lowering average premiums for the majority comes at the direct expense of a minority (e.g., chronically ill customers priced out of coverage). How should that tension be resolved?
4. What does a pure cost-benefit calculation leave out here? Is there a harm in this case that resists being converted into a number at all?

Part 2: From Consequences to Guardrails — Values and Norms

1. **Propose three norms** you would require for FairScore.
2. The CFO wants a business case for why these guardrails are worth slowing the rollout. **Build a chain of justification** (at least four links) connecting responsible underwriting to something that needs no further justification. **Identify which link is the intrinsic value and explain why it requires no further justification.**
3. Suppose it's legal in your state for insurers to use zip-code data in pricing, as long as race isn't an explicit input. Does legality settle the ethical question, given that zip code can closely track race and income? Use the legal-vs-moral norm distinction to explain.
4. The wellness-tracker discount is optional, but customers who skip it pay more by default. Is this "voluntary" in any meaningful sense, or is it a form of persuasive design that constrains choice — similar to a seatbelt-beep, but with money instead of sound? What obligation does that create for Halcyon's designers?

Part 3: When Guardrails Fail — Accountability

Suppose a customer with well-managed diabetes is denied a claim after FairScore flags her as high-risk, and it later emerges her risk score was inflated by an outdated prescription record the system never rechecked.

1. Place FairScore on the **automated–autonomous continuum**. Does a human underwriter meaningfully review each denial, or does the human role amount to a rubber stamp? If Halcyon removes human review entirely to cut costs, how does responsibility shift — does it disappear, or does it move somewhere else?
2. FairScore involves a chain of actors: the vendor who built the risk model, the actuarial team that tuned it for Halcyon, the executives who approved company-wide rollout, and the claims adjuster who denied this specific claim. **How would you distribute moral responsibility** across these actors for this wrongful denial? Could each actor plausibly claim only partial responsibility, such that no one is fully accountable — and what's dangerous about that outcome?
3. Before rollout, a data scientist testing FairScore noticed it denied claims at a higher rate for chronic-condition patients even after controlling for actual cost, but assumed the actuarial team would catch it in review. Using the two conditions for moral responsibility (moral autonomy; the ability to evaluate consequences), is the data scientist responsible for the harm that followed? How does this compare to an actuary who knowingly tunes the model to deny borderline claims because it improves quarterly numbers?

Part 4: Closing Synthesis

Write a **150–200 word statement** to Halcyon's board defending your final recommendation on FairScore — drawing on at least one idea from each part above: the utilitarian trade-off you weighed, the norms you'd require, and where accountability should sit if the system fails a customer.